POWER BI PROJECT

UBER TRIP ANALYSIS DASHBOARD

Project documentation & implementation plan

OVERVIEW • TIME ANALYSIS • LOCATION INSIGHTS • TRIP DETAILS

PROJECT NOTES & IMPLEMENTATION PLAN

01 | Project Overview

Project purpose, key performance indicators, and expected value

Analyze Uber trip data in Power BI to understand booking trends, revenue, and trip efficiency—supporting data-driven operational decisions.

Total Bookings
Number of trips booked in the selected period.

Total Booking Value
Total revenue generated across bookings.

Average Booking Value
Average revenue per booking.

Total Trip Distance
Combined distance covered by all trips.

Average Trip Distance
Average distance travelled per trip.

Average Trip Time
Average trip duration.

Expected Outcomes

- Identify trends in bookings and revenue generation.
- Analyze trip efficiency in terms of distance and duration.
- Compare booking values and trip patterns across time periods.
- Provide insights that may help optimize pricing models and customer satisfaction.

02 | Dashboard 1: Overview Analysis

Core performance measures, booking patterns, and vehicle comparison

Dynamic Measure Selector

Use a disconnected selector table to switch visual values between Total Bookings, Total Booking Value, and Total Trip Distance.

Booking Breakdowns

Analyze bookings by payment type (such as card, cash, and wallet) and by trip period (day versus night).

Vehicle-Type KPI Grid

Use a table or matrix to compare Total Bookings, Total Booking Value, Average Booking Value, and Total Trip Distance by vehicle type.

Daily Booking Trend

Track daily trip-volume fluctuations, identify peak and off-peak days, and explore possible effects of holidays, events, or weather.

Additional Enhancements

- Dynamic chart title: Update the title to reflect the selected measure.
- Slicers: Provide filters for date, city, and other relevant fields.
- Tooltips: Show supporting details such as average booking value or trip distance.

Expected value: identify booking and revenue trends, compare trip patterns over time, and surface opportunities to improve pricing and customer experience.

03 | Location Analysis

Understand where trips begin and end, and how demand varies by location

Frequent Pickup Point

Find the most common trip starting locations to help plan driver availability and dynamic pricing.

Frequent Drop-off Point

Identify the most common destinations. Use the appropriate inactive relationship in the Power BI model when needed.

Farthest Trip

Find the longest trip by distance to investigate outliers, long-distance demand, and fare patterns.

Top 5 Locations by Bookings

Rank the five locations with the highest trip volume to support demand forecasting and resource planning.

Preferred Vehicle by Pickup

Identify the most frequently booked vehicle type at each pickup location to understand local customer preferences.

Modeling note

Use location IDs and trip pickup/drop-off fields carefully when setting up model relationships.

04 | Interactivity & Usability Enhancements

Features intended to make the report easier to explore and maintain

Data Details Bookmark

Display definitions of key metrics, table descriptions, data source, and refresh frequency.

Clear Slicers Button

Add a reset action so users can clear selections in one click.

Download Raw Data

Allow export to CSV or Excel using available Power BI export functionality; Power Automate may be considered where appropriate.

Implementation note

Test each enhancement in Power BI before describing it as a completed feature. The source notes describe these as implementation ideas.

05 | Dashboard 2: Time Analysis

Explore demand across time intervals to support operations and driver availability

A global dynamic measure selector should update relevant visuals between Total Bookings, Total Booking Value, and Total Trip Distance.

Pickup Time | Area Chart

Group bookings into 10-minute intervals across the day to identify demand peaks and quieter periods.

Day Name | Trend Chart

Compare booking patterns across days of the week. The source notes specify Monday to Saturday; use the days present in your data.

Hour × Weekday | Heatmap

Matrix rows: hours 0–23. Columns: Monday–Sunday. Values: selected dynamic measure. Use the heatmap to spot recurring demand peaks.

Business use

Align driver availability and operational planning with the times and days that show the highest demand.

06 | Dashboard 3: Details Tab

Provide granular trip records with contextual drill-through

Grid Table

Display essential trip fields so users can inspect records behind aggregated charts.

Drill-through

Allow users to right-click a point in a supported chart or heatmap and open the details tab filtered to that selection.

Full-Data Bookmark

Provide a “View Full Data” bookmark to switch between filtered drill-through results and the complete dataset.

Suggested user journey

- Select a chart or heatmap data point.
- Choose the Details tab through drill-through.
- Review the related trip records.
- Use the full-data bookmark or reset filters to return to all records.

07 | Data Model & Measures

Relationships, date analysis, and core DAX measures from the project notes

Relationships and date fields

- Location ID to Location ID (many-to-one relationship).
- Trip drop-off location to the location ID table.
- Check active/inactive relationship behavior for pickup and drop-off analysis.
- Create a Pickup Date column.
- Create a Calendar table to support time-based filtering and analysis.

Core Measures

Measure	Definition
Total Bookings	Count of Trip ID
Total Booking Amount	Sum of Fare Amount + Sum of Surge Fee
Average Booking Amount	DIVIDE([Total Booking Amount], [Total Bookings], BLANK())
Total Trip Distance	Sum of Trip Distance
Average Trip Distance	Average of Trip Distance
Average Trip Time	Average duration in minutes, calculated from pickup and drop-off times.

Validation note: confirm the table and column names against the actual Power BI model before reusing DAX expressions.

08 | Implementation Checklist

A practical sequence for completing and presenting the project

1. Prepare data	Confirm trip, location, and date fields; create the calendar table and relationships.
2. Build core KPIs	Create and validate booking, booking value, distance, and duration measures.
3. Create overview visuals	Add dynamic measure selection, payment/trip breakdowns, vehicle matrix, and daily booking trend.
4. Add time analysis	Build 10-minute pickup intervals, day-of-week trends, and hour-by-weekday heatmap.
5. Add details navigation	Configure drill-through, the detailed grid, and the full-data bookmark.
6. Polish and document	Test slicers and tooltips, take page screenshots, and document only features that are implemented.

Project goal

Turn trip-level data into a clear, interactive report that helps users explore booking patterns, revenue, trip efficiency, and location demand.